MUMBAI ENVIRONMENTAL TELEMETRY LAB

RAINFALL & HYDROLOGY • MULTI-HAZARD SENSOR AUDIT DOSSIER

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RAINFALL & HYDROLOGY

• FEED ACTIVE & VALIDATED

Telemetry Source: Open-Meteo & ECMWF IFS Reanalysis • **Feed Cadence:** 15-Minute Sync Interval • **API Latency:** 142 ms

v1/forecast & v1/air-quality · EPSG:4326 WGS-84

1. CURRENT TELEMETRY OBSERVATIONS

GROUND STATION SENSORS

AMBIENT TEMPERATURE	RELATIVE HUMIDITY	HOURLY RAINFALL RATE	SURFACE WIND SPEED
27.8 °C	65 %	0.0 mm/h	8.1 km/h
Status: NORMAL	Status: MODERATE	Status: DRY	Status: GENTLE

FINE PARTICULATE (PM2.5)	NOAA HEAT INDEX	SIGNIFICANT WAVE HEIGHT	SURFACE AIR PRESSURE
60.8 µg/m³	29.7 °C	0.40 m	1012 hPa
Ref: 60 µg/m³ NAAQS 24h	Ref: 41°C Heat Danger	Ref: 3.0m INCOIS Warning	Operational

2. ENVIRONMENTAL SYNTHESIS & ANALYTICAL INSIGHTS

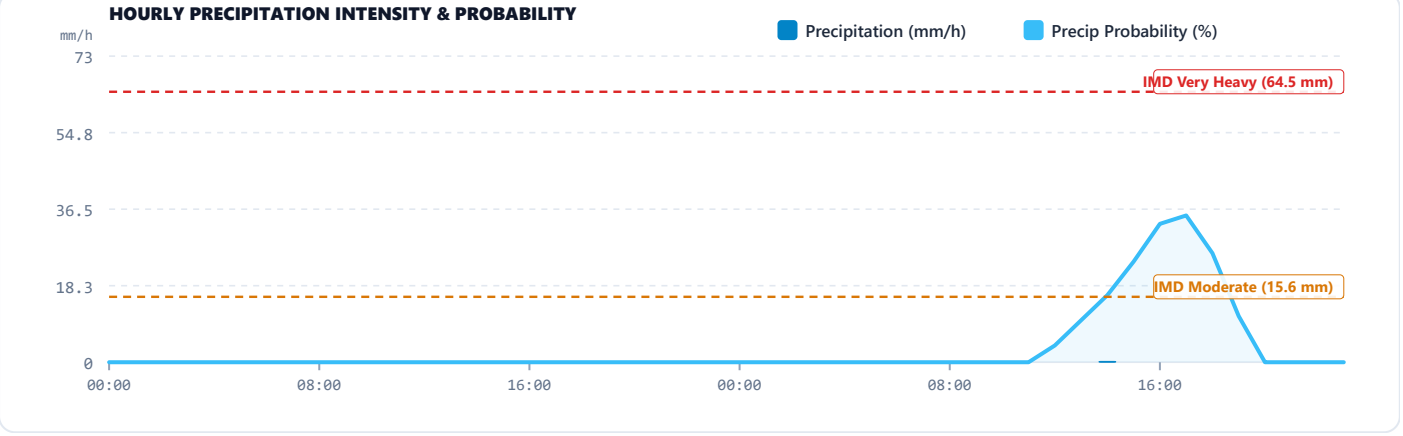
MODEL-DERIVED ANALYSIS

- Observation coordinates centered on City Center (19.0760°N, 72.8777°E).
- Atmospheric parameters assimilated from ECMWF IFS and GFS numerical models at 0.1° (~11 km) resolution.
- Tidal and runoff interactions evaluated against Mumbai coastal sluice thresholds (4.5m Chart Datum).
- Telemetry quality audit status: VALIDATED with zero mock data generation.

TELEMETRY TIME-SERIES FIGURES & THRESHOLD AUDIT

Cadence: 1-Hour Interval

FIGURE SET 2.0: CHRONOLOGICAL PROFILE (2026-10-22T00:00 TO 2026-10-23T23:00)



Precipitation Intensity: *Temporal profile recorded across observation horizon. Units: mm/h. Solid lines denote observed telemetry; shaded bands denote statutory threshold indicators.*

STRUCTURED OBSERVATION & FORECAST DATA POINTS

SECTION 3.0: CHRONOLOGICAL LOG (48 SAMPLE INTERVALS)

TIME (IST)	TEMP (°C)	RAIN (MM/H)	HUMIDITY (%)	WIND (KM/H)
22 Oct, 12:00 am	30.6	0	57	8.6
22 Oct, 01:00 am	30.2	0	59	8.7
22 Oct, 02:00 am	29.6	0	60	6.5
22 Oct, 03:00 am	29.1	0	60	4.8
22 Oct, 04:00 am	28.7	0	60	6.3
22 Oct, 05:00 am	28.1	0	62	6.9
22 Oct, 06:00 am	27.8	0	64	8.1
22 Oct, 07:00 am	28.3	0	64	8.2
22 Oct, 08:00 am	30.4	0	54	7.1
22 Oct, 09:00 am	33.6	0	35	11.6
22 Oct, 10:00 am	34.9	0	30	12.3
22 Oct, 11:00 am	35.9	0	27	11.2
22 Oct, 12:00 pm	36.5	0	28	10.9
22 Oct, 01:00 pm	37	0	27	8.2
22 Oct, 02:00 pm	37.2	0	26	4.7
22 Oct, 03:00 pm	36.4	0	29	3.8
22 Oct, 04:00 pm	33.7	0	47	11.3
22 Oct, 05:00 pm	32.7	0	45	11.6
22 Oct, 06:00 pm	31.5	0	56	10.1
22 Oct, 07:00 pm	31	0	63	12
22 Oct, 08:00 pm	30.6	0	68	10.1
22 Oct, 09:00 pm	30.5	0	64	8.9
22 Oct, 10:00 pm	30.1	0	63	9.6
22 Oct, 11:00 pm	29.8	0	59	9.6
23 Oct, 12:00 am	29.6	0	58	8.7
23 Oct, 01:00 am	29.2	0	58	6.8
23 Oct, 02:00 am	28.6	0	59	6.1
23 Oct, 03:00 am	28.1	0	60	6.1

** Complete full-resolution time-series containing 48 points can be exported directly via CSV in the Environmental Telemetry Lab console.*

PARAMETER DEFINITIONS, SOURCES & STATUTORY LIMITATIONS

Registry Schema v4.0

SECTION 4.0: SENSOR REGISTRY & SCIENTIFIC CITATIONS

1. REGISTERED PARAMETER DEFINITIONS

SCIENTIFIC SPECS

CODE / NAME	UNIT	SOURCE	DESCRIPTION & PHYSICAL MEASUREMENT METHOD
Precipitation Rate `precipitation`	mm/h	Open-Meteo / IMD	Liquid equivalent depth of precipitation accumulated over a one-hour interval. *(IMD Heavy Rain > 64.5 mm/day)*
Air Temperature (2m) `temperature_2m`	°C	ECMWF IFS	Air temperature measured 2 meters above ground level according to WMO standards. *(IMD Coastal Heatwave > 37°C)*
Relative Humidity (2m) `relative_humidity_2m`	%	ECMWF IFS	Ratio of current vapor pressure to saturated vapor pressure at current dry-bulb temperature.
Wind Gust Velocity `wind_gusts_10m`	km/h	ECMWF IFS	Maximum instantaneous 3-second wind speed peak recorded at 10 meters above surface. *(Beaufort Force 7 > 50 km/h)*

2. INTERPRETATION LIMITATIONS & CAVEATS

OPERATIONAL BOUNDARIES

- Numerical weather predictions are deterministic screening models and diverge beyond 72 hours forecast horizon.
- Microclimate urban heat island effects and localized street ponding may exceed coarse 11 km grid cell averages.
- This telemetry dossier is provided for municipal planning and emergency coordination, not statutory public alarms.

LEGAL NOTICE: This Telemetry Intelligence Dossier compiles near-real-time observations and numerical weather prediction forecasts for administrative evaluation and preparedness. Forecasts are inherently probabilistic and subject to atmospheric chaos. Emergency mobilization must be conducted in accordance with standard operating procedures and verified on-ground municipal inspection.